FIITJEE

ALL INDIA TEST SERIES

FULL TEST – VII

JEE (Advanced)-2020

PAPER – 1

Time Allotted: 3 Hours **Maximum Marks: 183**

General Instructions:

- The test consists of total 54 questions.
- Each subject (PCM) has 18 questions.
- This question paper contains **Three Parts.**
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each **Part** is further divided into **Two Sections**: **Section-A & Section-B.**

1. ***Section-A (01– 7, 19 – 25, 37 – 43)*** *contains 21 multiple choice questions which have* ***one or more than one correct*** *answer. Each question carries* ***+4 marks*** *for correct answer and* ***–2 marks*** *for wrong answer*

   Partial Marks **+1** for each correct option provided no incorrect option is selected.

   Section-A (08 – 13, 26 – 31, 44 – 49) *contains 18 questions. Each of 2 Tables with 3 Columns and 4 Rows has three questions. Column 1 will be with 4 rows designated (I), (II), (III) and (IV). Column 2 will be with 4 rows designated (i), (ii), (iii) and (iv). Column 3 will be with 4 rows designated (P), (Q), (R) and (S). Each multiple choice question with 4 options has* ***only one correct*** *answer and carries* ***+3 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

2. ***Section–B (14 – 18, 32 – 36, 50 – 54)*** *contains 15 Numerical based questions with answers as numerical value from 0 to 9 and each question carries* ***+3 marks*** *for correct answer. There is no negative marking.*

Physics

PART – I

SECTION – A
(One or More than one correct type)

This section contains **07** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

1. Nucleus A decays to B with decay constant λ_1 and B decays constant λ_2. Initially at t = 0, number of nuclei of A and B are $2N_0$ and N_0 respectively. At $t = t_0$, number of nuclei of B stop changing. If at this instant number of nuclei of B are $\frac{3N_0}{2}$.

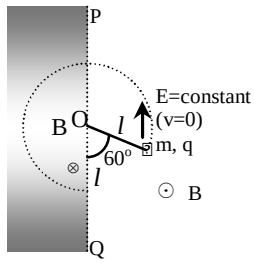
   (A) The value of t_0 is $\frac{1}{\lambda_1}\ln\frac{4}{3}\frac{\lambda_1}{\lambda_2}$

   (B) The value of t_0 is $\frac{1}{\lambda_2}\ln\frac{4}{3}\frac{\lambda_1}{\lambda_2}$

   (C) The value of N_A at t_0 is $\frac{3N_0}{2}\frac{\lambda_2}{\lambda_1}$

   (D) The value of N_A at t_0 is $\frac{2N_0}{3}\frac{\lambda_2}{\lambda_1}$

2. Consider a bob of mass m and having charge q attached with a light string of length l and pivoted at point O. It is released at rest at $60°$ with vertical. There are two regions– Region-I (left of line PQ) has a uniform and constant magnetic field B directed inside plane of paper. Region-II (right of line PQ) has a constant and uniform electric field E directed vertically up as shown. Consider no effect of gravity in both the regions.



   (A) Time taken by particle to cross region-I for 1st time is $\pi\sqrt{\frac{2ml}{5qE}}$

   (B) Time taken by particle to cross region-I for 2nd time is $\pi\sqrt{\frac{2ml}{13qE}}$

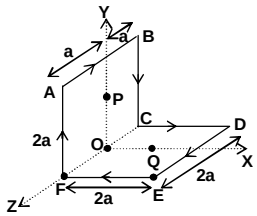
   (C) Angular speed of 1st revolution in magnetic field is $\sqrt{\frac{5qE}{2ml}}$

   (D) Angular speed of 2nd revolution in magnetic field is $\sqrt{\frac{13qE}{ml}}$

Space for Rough work

3. A binary stars system in which stars are considered as point mass having mass m and 2m are separated by a large distance 'r'. Choose the correct statement(s)
(A) Radius of circular path of star of mass 2m is r/3.
(B) Kinetic energy of 2m mass star is one forth of that of lighter star.
(C) Time period of revolution of each star is same
(D) Angular momentum of lighter star is more.

4. A wire frame ABCDEF is kept as shown in figure. I am the current in the wire frame.



(A) The direction of magnetic field at 'O' due to the wire frame will be along $\left(\frac{-\hat{i}-\hat{j}}{\sqrt{2}}\right)$

(B) The direction of magnetic field at P (0, a, 0) due to wire CD and EF will be along $-\hat{j}$

(C) The direction of magnetic field at Q(a, 0, 0) due to wire EF and CD will be along $\left(-\hat{j}\right)$

(D) The direction of magnetic field at P(0, a, 0) due to wire AF and CB will be along $\left(-\hat{i}\right)$

5. Which of the following represents a progressive harmonic wave?
(A) A (ax – bt)
(B) A tan (ωt – kx)
(C) A cos (vt – x)
(D) A cos kx sinωt

6. A small ball of mass 'm'& radius r is released from rest in a viscous liquid of coefficient of viscosity η and density half of that of ball. Choose the correct option(s)
(A) The ball will never attain constant velocity

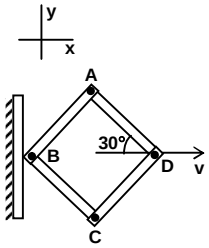
(B) After time $t=\frac{m\ell n2}{6\pi\eta r}$ the velocity of ball will be half of the terminal velocity

(C) The instantaneous velocity of ball at a time t is given by $v=v_T\left(1-e^{-\frac{6\pi\eta r}{m}t}\right)$

(D) The velocity of ball will remain constant

Space for Rough work

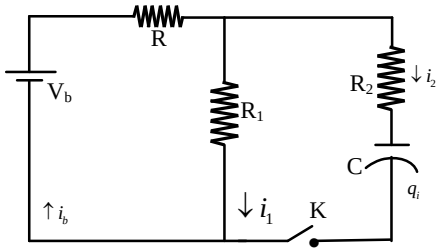
7. ABCD is a wire frame of identical wires in which point D is given velocity v as shown in figure. Choose the correct statement(s)
(A) Velocity of point A along x-axis will be v/2
(B) Speed of point A will be v
(C) Speed of point A along y-axis will v/2
(D) Velocity of point A will be equal to velocity of point C



(Matching type - Single Correct Option)

This section contains **SIX** questions of matching type. The section contains **TWO** tables (each having 3 columns and 4 rows). Based on each table, there are **THREE** questions. Each question has **FOUR** options (A), (B), (C), and (D). **ONLY ONE** of these four options is correct.

Answer questions Q. 8, Q. 9 and Q. 10 by appropriately matching the information given in the three columns of the following table.



q_i is the initial charge on the capacitor. The key K is closed at t=0

Column 1		Column 2		Column 3	
(I)	$V_b = 0V$	(i)	$R_1 = R_2 = 5\Omega$ $R = 10\Omega$	(P)	$C = 5\mu F$ $q_i = 0$
(II)	$V_b = 30V$	(ii)	$R_1 = 5\Omega$ $R_2 = 10\Omega$ $R = 10\Omega$	(Q)	$C = 20\mu F$ $q_i = 0$
(III)	$V_b = -30V$	(iii)	$R_1 = 10\Omega$ $R_2 = 5\Omega$ $R = 10\Omega$	(R)	$C = 5\mu F$ $q_i = 100\mu C$
(IV)	$V_b = 60V$	(iv)	$R_1 = R_2 = R = 10\Omega$	(S)	$C = 20\mu F$ $q_i = 100\mu C$

Space for Rough work

8. In which of the following cases, will the current $i_1 = \frac{5}{3}\text{A}$, initially?

(A) $(\text{I})\ (\text{i})\ (\text{P})$

(B) $(\text{II})\ (\text{iv})\ (\text{R})$

(C) $(\text{II})\ (\text{iv})\ (\text{P})$

(D) $(\text{I})\ (\text{iii})\ (\text{Q})$

9. In which of the following cases will the net charge flowing through the capacitor be equal to 300 μC ?

(A) $(\text{II})\ (\text{i})\ (\text{S})$

(B) $(\text{I})\ (\text{ii})\ (\text{R})$

(C) $(\text{IV})\ (\text{i})\ (\text{S})$

(D) $(\text{III})\ (\text{iii})\ (\text{P})$

10. In which case will the current through R decay exponentially to zero with a time constant of 0.2 ms?

(A) $(\text{IV})\ (\text{ii})\ (\text{Q})$

(B) $(\text{I})\ (\text{iii})\ (\text{S})$

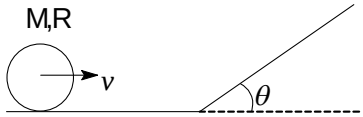
(C) $(\text{II})\ (\text{i})\ (\text{R})$

(D) $(\text{I})\ (\text{iii})\ (\text{Q})$

Space for Rough work

Answer questions Q. 11, Q. 12 and Q. 13 by appropriately matching the information given in the three columns of the following table.

An object of mass M and radius R is pure rolling on a horizontal surface. It smoothly climbs over an rough incline with sufficient friction to ensure no slipping. The following grid has 3-columns: Column- I represents the type of object. Column II represents the maximum height attained by the object and column III represents the time taken to reach the maximum height.



Column 1 (Object)		Column 2 (Height)		Column 3 (Time)	
(I)	Ring	(i)	$\frac{7v^2}{10g}$	(P)	$7v/5g\sin\theta$
(II)	Disc	(ii)	$\frac{3v^2}{4g}$	(Q)	$2v/g\sin\theta$
(III)	Hollow sphere	(iii)	v^2/g	(R)	$3v/2g\sin\theta$
(IV)	Solid sphere	(iv)	$\frac{5v^2}{6g}$	(S)	$5v/3g\sin\theta$

11. Pick the correct combination:-

(A) $(\mathrm{I})\ (\mathrm{i})\ (\mathrm{P})$

(B) $(\mathrm{III})\ (\mathrm{ii})\ (\mathrm{Q})$

(C) $(\mathrm{II})\ (\mathrm{ii})\ (\mathrm{R})$

(D) $(\mathrm{IV})\ (\mathrm{iv})\ (\mathrm{P})$

Space for Rough work

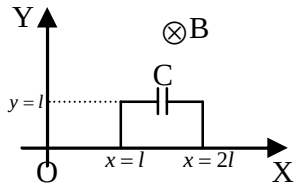
12. The combination where the total mechanical energy is maximum will be
(A) $(\mathrm{I})\ (\mathrm{iii})\ (\mathrm{Q})$
(B) $(\mathrm{II})\ (\mathrm{ii})\ (\mathrm{R})$
(C) $(\mathrm{III})\ (\mathrm{iv})\ (\mathrm{P})$
(D) $(\mathrm{IV})\ (\mathrm{i})\ (\mathrm{S})$

13. The combination for which maximum friction acts between the body and the incline is
(A) $(\mathrm{I})\ (\mathrm{iii})\ (\mathrm{Q})$
(B) $(\mathrm{II})\ (\mathrm{ii})\ (\mathrm{R})$
(C) $(\mathrm{III})\ (\mathrm{iv})\ (\mathrm{S})$
(D) $(\mathrm{IV})\ (\mathrm{i})\ (\mathrm{P})$

SECTION – B
(Single digit integer type)

This section contains **FIVE** questions. The answer to each question is a single Digit integer ranging from 0 to 9, both inclusive.

14. A square frame of wire connected to a capacitor is kept in a magnetic field which varies with position as well as time and is given as $\vec{B}=\alpha xt\left(-\hat{k}\right)$ as shown in figure, where α is a constant. The charge on the capacitor at any time is $\frac{n}{2}C\alpha l^3$. Then find the value of 'n'.



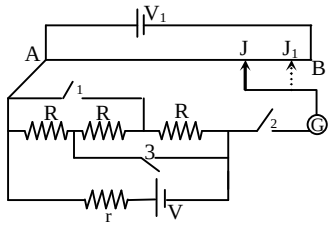
Space for Rough work

15. A large concave mirror with a radius of curvature of 0.5 m is placed on a horizontal plate. We pour a small amount of water into the mirror and place on it a plano-convex lens with a radius of curvature of 1.0 m whose plane side just covers the water. Then we place a point light source above this set-up at a height of 0.3 m. The distance (in metres) from the object where the final image is formed will be $\frac{x}{90}$. Find x given that the refractive index of water is $\frac{4}{3}$ and the refractive index of glass is 1.50.

16. A radioactive sample contains two nuclides A and B having decay constant λhr^{-1} and $2\lambda hr^{-1}$. Initially 25% of total decay comes from A. How long (in hr) will it take before 75% of total decay comes from A.
[Take $\lambda = \ln 3$]

17. An electron accelerated by a potential difference $V = 3.6\ \text{V}$, first enters a region of uniform electric field of a parallel-plate capacitor whose plates extend over a length $l = 6\ \text{cm}$ in the direction of initial velocity. The electric field is normal to the direction of initial velocity and its strength varies with time as $E = a \times t$ where $a = 3200\ \text{Vm}^{-1}\text{s}^{-1}$. Then the electron enters a region of uniform magnetic field of induction $B = \pi \times 10^{-9}\ \text{T}$. Direction of magnetic field is same as that of the electric field. Calculate the pitch (in mm) of helical path traced by the electron in the magnetic field. (Mass of electron, $m = 9 \times 10^{-31}\ \text{kg}, e = 1.6 \times 10^{-19}\ \text{C}$) .
[Neglect the effect of induced magnetic field].

18. In the given circuit wire AB is of uniform resistance. On closing switch 2 with other two switches open, the balance length AJ is L_1. On closing all the switches, the balance length AJ_1 is L_2. The internal resistance of the battery V is $nR\Omega$.Find n. $\left(\frac{L_1}{L_2} = 3\right)$



Space for Rough work

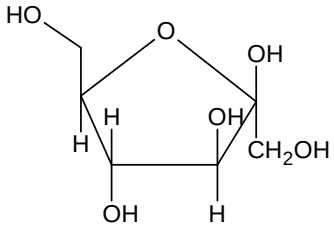
Chemistry

PART – II

SECTION – A
(One or More than one correct type)

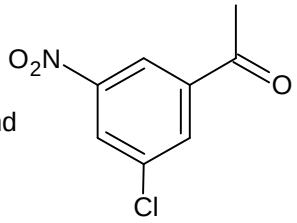
This section contains **07** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

19. The correct statement regarding compound is:

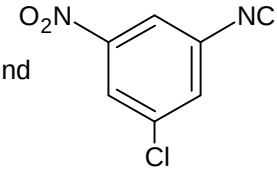


(A) Can reduce Tollen's reagent
(B) It is furanose compound
(C) It can show mutarotation
(D) Stable in basic condition

20. An oxime of compound



whenever treated with H_2SO_4/Δ gives a compound P, which on hydrolysis produces Q and R. If Q is an aliphatic primary amine, then choose the correct option(s)?
(A) R is soluble in hot aqueous NaOH
(B) R gives brisk effervence of CO_2 with aqueous $NaHCO_3$
(C) Q can give foul smelling compound



with $CHCl_3$/NaOH
(D) Q on Hofmann exhaustive methylation gives product which has very high solubility in water

Space for Rough work

21. $Zn(OH)_2(s) \rightleftharpoons Zn(OH)_2(aq.)$ $K_1 = 10^{-6}\ M$

$Zn(OH)_2(aq.) \rightleftharpoons Zn(OH)^+ + OH^-$ $K_2 = 10^{-7}\ M$

$Zn(OH)^+ \rightleftharpoons Zn^{2+} + OH^-$ $K_3 = 10^{-4}\ M$

$Zn(OH)_2(aq.) + OH^- \rightleftharpoons Zn(OH)_3^-$ $K_4 = 10^3\ M^{-1}$

$Zn(OH)_3^- + OH^- \rightleftharpoons Zn(OH)_4^{2-}$ $K_5 = 10\ M^{-1}$

Which of the following options are correct about solubility of zinc hydroxide at various pH.

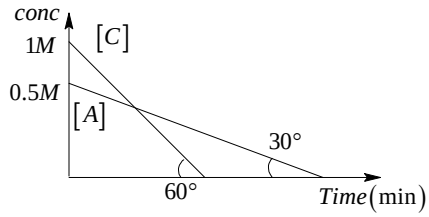
(A) $2\times10^{-4}\ M\ at\ pH = 13$

(B) $1.12\times10^{-6}\ M\ at\ pH = 9$

(C) $10\ M\ at\ pH = 5$

(D) $10^{-10}\ M\ at\ pH = 5$

22. For the two reactions $I: A \rightarrow B; II: C \rightarrow D$ following graph is obtained:



Which of the following statement is false?

(A) If $[B] = [A]$ then at that time $[D] = 0.75M$

(B) If $[C] = [A]$ then at that time $[B] > [D]$

(C) $(t_{100\%})_{\text{Reaction I}} = (t_{100\%})_{\text{Reaction II}}$

(D) $[A] = [C]$ at $t = \dfrac{\sqrt{3}}{2}$ min

23. The ions for which all octahedral complexes are paramagnetic irrespective of ligand?

(A) Sc^{3+}, Ti^{3+}

(B) Cr^{3+}, Co^{3+}

(C) Ti^{3+}, V^{3+}

(D) Fe^{3+}, Ni^{3+}

Space for Rough work

24. In the reaction sequence given below A is an anionic Fe(II) complex

$$A \xrightarrow{aq.NO_2^-} \underset{\text{brown}}{B} \xrightarrow{aq.S^{-2}} \underset{(\text{Purple})}{C}$$

Match the A,B, C with correct option.

	COLUMN – I		**COLUMN-II**
(1)	A	(P)	$Na_2[Fe(CN)_5NO]$
(2)	B	(Q)	$Na_4[Fe(CN)_6]$
(3)	C	(R)	$Na_4[Fe(CN)_5(NO)(SO_3)]$
		(S)	$Na_4[Fe(CN)_5NOS]$
		(T)	$[Fe(NO)(H_2O)_5]SO_4$
		(U)	Hybridisation of central metal atom is d^2sp^3
		(V)	Sodium penta cyano nitrosyl ferrate (II)

(A) 1-QU, 2-PSR, 3-SU
(B) 1-QU, 2-PUV, 3-PT
(C) 1-QV, 2-PUV, 3-SU
(D) 1-QU, 2-PUV, 3-SU

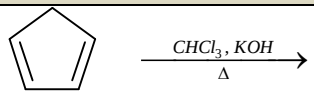
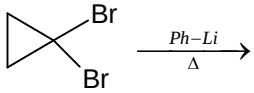
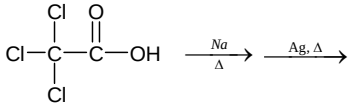
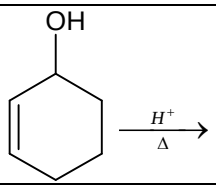
25. In a reaction $[Co(NH_3)_6]^{2+} \xrightarrow[\text{air}]{\text{bubble in}} [Co(NH_3)_6]^{3+}$, which of the following are correct about this reaction?
(A) Co^{3+} complex is more stable then Co^{2+} complex
(B) Charge transfer spectra is responsible for colour of both complexes
(C) Reduction by air
(D) Co^{3+} has high crystal field stabilization energy than Co^{2+}.

Space for Rough work

(Matching type - Single Correct Option)

This section contains **SIX** questions of matching type. The section contains **TWO** tables (each having 3 columns and 4 rows). Based on each table, there are **THREE** questions. Each question has **FOUR** options (A), (B), (C), and (D). **ONLY ONE** of these four options is correct.

Answer questions Q. 26, Q. 27 and Q. 28 by appropriately matching the information given in the three columns of the following table.

	Column 1 (Reaction)		Column 2 (Intermediate)		Column 3 (Double bond equivalent in the Final product)
(I)		(i)	Carbocation	(P)	0
(II)		(ii)	Carbanion	(Q)	4
(III)		(iii)	Free Radical	(R)	2
(IV)		(iv)	Carbene	(S)	3

26. Which of the following is correct combination if the given matrix?

(A) $(\mathrm{II})\ (\mathrm{iv})\ (\mathrm{R})$

(B) $(\mathrm{I})\ (\mathrm{ii})(\mathrm{iv})\ (\mathrm{R})$

(C) $(\mathrm{II})\ (\mathrm{ii})(\mathrm{iv})\ (\mathrm{R})$

(D) $(\mathrm{III})\ (\mathrm{iii})\ (\mathrm{P})$

Space for Rough work

27. Which of the following is correct combination if the given matrix?

(A) $(IV)(i)\ (R)$

(B) $(I)\ (iv)\ (R)$

(C) $(III)\ (iv)\ (P)$

(D) $(I)\ (ii)(iv)\ (Q)$

28. In which case the product is used for anesthetic purposes in old time

(A) $(IV)(i)\ (R)$

(B) $(IV)\ (i)\ (Q)$

(C) $(III)\ (ii)\ (R)$

(D) $(III)\ (iv)\ (Q)$

Answer questions Q. 29, Q. 30 and Q. 31 by appropriately matching the information given in the three columns of the following table.

Column-1 [Reactant-1]		Column-2 [Reactant-2]		Column-3 [Product]	
(I)	MnO_4^-	(i)	$H_2C_2O_4$	(P)	MnO_2
(II)	$K_2Cr_2O_7$	(ii)	$Na_2S_2O_3$ in H^+	(Q)	$MnSO_4$
(III)	I_2	(iii)	SO_2	(R)	CO_2
(IV)	CO	(iv)	Fe_2O_3	(S)	H_2SO_4

29. Which of the following combinations is not possible?

(A) $(I)\ (i)\ (R)$

(B) $(II)\ (iii)(S)$

(C) $(III)\ (ii)(S)$

(D) $(IV)\ (iv)\ (R)$

Space for Rough work

30. Which of the following is not possible in an acidic medium?
(A) $(\mathrm{I})\ (\mathrm{iv})\ (\mathrm{P})$
(B) $(\mathrm{II})\ (\mathrm{iii})(\mathrm{S})$
(C) $(\mathrm{I})\ (\mathrm{iii})(\mathrm{Q})$
(D) $(\mathrm{II})\ (\mathrm{i})\ (\mathrm{R})$

31. In which of the following combinations will the reactants react in 2:5 ratio.
(A) $(\mathrm{II})\ (\mathrm{i})\ (\mathrm{R})$
(B) $(\mathrm{I})\ (\mathrm{i})(\mathrm{Q})$
(C) $(\mathrm{IV})\ (\mathrm{iv})(\mathrm{R})$
(D) $(\mathrm{I})\ (\mathrm{iii})\ (\mathrm{S})$

SECTION – B
(Single digit integer type)

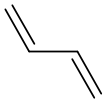
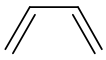
This section contains **FIVE** questions. The answer to each question is a single Digit integer ranging from 0 to 9, both inclusive.

32. Predict whether or not AgCl will be precipitated from a solution which is $0.02\mathrm{M}$ in NaCl and $0.1\mathrm{M}$ in $[\mathrm{Ag}(CN)_2]^-$. Given $K_{\text{dissociation of the complex ion}} = 4.0\times10^{-17}$ and $K_{sp}\ \mathrm{AgCl}=2.8\times10^{-10}$.
(i) mark 0, if ppt of $AgCl$ form
(ii) mark 1, if no ppt form
(iii) mark 2, if ppt of $NaCN$ form
(iv) mark 3, if ppt of KCN form

33. A $0.5\,mL$ sample of an aqueous solution containing $0.48\,\mu Ci$ per mL of Na^{22} is injected into the bloodstream of an animal. After allowing sufficient time for complete circulatory mixing a $0.1\,mL$ sample of blood is removed and found to have a net activity of $444\ dpm$ of Na^{22} from this data, the blood volume of animal was found to be $60x$ mL. Find the value of x.

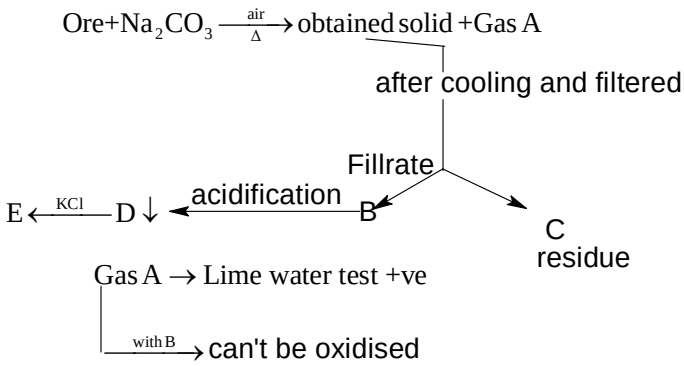
Space for Rough work

34. For the two parallel reaction $A \xrightarrow{k_1=2\sec^{-1}} B$ and $A \xrightarrow{k_2=4\sec^{-1}} C$, the activation energy E' for the disappearance of A is given in terms of activation energies E_1 and E_2 for the two path by
$E' = \frac{E_1 + 2E_2}{x}$. The value of x is…….

35. The correct statement(s) is/are 'x' then value of $\frac{x}{2.5}$ will be:
(a) In some cases, constitutional isomers are chiral
(b) In every cases, a pair of enantiomers have a mirror-image relationship
(c) Mirror-image molecules are in all cases enantiomers
(d) If a compound has an enantiomer it must be chiral
(e) Every chiral compound has a diastereomer
(f) If a compound has a diastereomer it must be chiral
(g) Every molecule containing one or more asymmetric carbons is chiral
(h) Any molecule containing a stereocentre must be chiral
(i) Any molecule with a stereocentre must show stereoisomerism
(j) Some diastereomers are mirror images to each other
(k) Any chiral compound with a single asymmetric carbon must have a positive optical rotation if the compound has the R configuration.
(l) If a structure has no plane of symmetry it is chiral
(m) All chiral molecules have no plane of symmetry
(n) All asymmetric carbons are stereocentre
(o) Gauche form of n-butane is chiral however anti form is achiral
(p) Molecules with two or more identically substituted stereocenters may exists as meso stereoisomers
(q) A stereocentre can be tetrahedral or trigonal planer
(r)  and  both are comformational diastereomers.

Space for Rough work

36.



E is an oxidising agent and when a paper dipped in E solution comes in contact with SO_2 changes colour from orange to green.

Now, how many are correct about above flow diagram?

(A) Lime is added with Na_2CO_3, keeps the mass porous so that air becomes accessible to all parts reaction mixture and prevents fusion

(B) 'C' is mainly Fe_2O_3

(C) Filtrate is a yellow solution.

(D) During acidification a redox reaction is taking place.

(E) D contain 2 salts in solution and when fractional crystallisation occurs salt of lower molecular mass comes out first and is filtered off.

(F) Acidication gives two salts both hydrated. The one which has more water of hydration is less soluble

(G) When pH of solution is increased the colour of solution of E changes from orange to yellow

(H) The oxidising anions present in flow diagram can be distinguished by $AgNO_3$ One gives reddish-brown ppt and other gives brick red ppt.

Space for Rough work

Mathematics

PART – III

SECTION – A
(One or More than one correct type)

This section contains **07** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

37. If the imaginary part of $\frac{z-3}{e^{1\theta}}+\frac{e^{i\theta}}{z-3}$ is zero, then z lies on
(A) A circle with unit radius
(B) A circle with radius 3 units
(C) A straight line through the point (3, 0)
(D) A parabola with the vertex (3, 0)

38. If $\frac{^{50}C_{5r}}{^{50}C_{r}}=29\times43\times\frac{a}{b}$ where $a\in I^{+}$ is not a multiple of 47 and $b\in I^{+}$ then which of the following is true?
(A) r is a prime number
(B) r is a composite number
(C) r has two different integer values
(D) exponent of prime number 3 in $\frac{^{50}C_{5r}}{^{50}C_{r}}$ is 2

39. If $\lim_{x\to0}\frac{\sin\left(m^2+n^2\right)x}{\tan\frac{x}{\sqrt{3}}}=25$ where m, n $\in$ N
(A) m = 4, n = 3
(B) n = 3, m = 4
(C) m = 5, n = 1
(D) m = 1, n = 5

Space for Rough work

40. Tangents are drawn to the circle $x^2 + y^2 = 50$ from a point 'P' lying on the x-axis. These tangents meet the y-axis at points 'P_1' and 'P2'. Possible coordinates of 'P' so that area of triangle PP_1P_2 is minimum, are
(A) (10, 0)
(B) $(10\sqrt{2}, 0)$
(C) (-10, 0)
(D) $(-10\sqrt{2}, 0)$

41. The chord x+y=1 cuts the curve $y^2=12x$ in points A and B. The normals at A and B intersect at C. A third line from C cuts the $y^2=12x$ normally at D. Then which of the following is/are correct
(A) Sum of slopes of normals at A and B is 3
(B) Coordinates of D are (12,12)
(C) Tangent at D intersect y axis at (0,6)
(D) Coordinates of D are (12,–12)

42. The parametric angle θ, where $-\pi < \theta \leq \pi$, of the point on the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2} = 1$ at which the tangent drawn cuts the intercept of minimum length on the co-ordinates axis, is/are:
(A) $\tan^{-1}\sqrt{\frac{b}{a}}$
(B) $-\tan^{-1}\sqrt{\frac{b}{a}}$
(C) $\pi - \tan^{-1}\sqrt{\frac{b}{a}}$
(D) $\pi + \tan^{-1}\sqrt{\frac{b}{a}}$

Space for Rough work

43. A chord AB, which is bisected at (1, 1) is drawn to the hyperbola $x^2 + 4xy - y^2 - 3 = 0$ with centre 'C' which intersects its asymptotes in E and F. Then

(A) Radius of the circum circle of Δ CEF is $\sqrt{2}$

(B) It e_k be the eccentricity of hyperbola $x^2 + 4xy - y^2 - 3k = 0 \ \forall k = 1, 2, 3, \ldots..2014$, then $\prod_{k=1}^{2014} e_k = 2^{1007}$

(C) Slope of the line $\perp$ to the segment AE is $\frac{1}{3}$

(D) All A, B, C cannot be true simultaneously

(Matching type - Single Correct Option)

This section contains **SIX** questions of matching type. The section contains **TWO** tables (each having 3 columns and 4 rows). Based on each table, there are **THREE** questions. Each question has **FOUR** options (A), (B), (C), and (D). **ONLY ONE** of these four options is correct.

Answer questions Q. 44, Q. 45 and Q. 46 by appropriately matching the information given in the three columns of the following table.

Let $f(x) = \frac{(x-a)(x-b)}{(x-c)(x-d)}$ where $a < b < c < d$. Column-1: contains information about real roots of the equation; Column-2: contains information about limit of $f(x)$; column-3: contains information about increasing interval and decreasing interval of $f(x)$ *and* $f'(x)$ and information about local minimum value and local maximum value of $f(x)$

	Column 1		Column 2		Column 3
(I)	$f(x) = 1$ has exactly one real root	(i)	$\lim_{x \to c^-} f(x) = -\infty$	(P)	$f(x)$ is increasing in the interval (b, c)
(II)	$f(x) = 2$ has exactly two real root	(ii)	$\lim_{x \to d^+} f(x) = \infty$	(Q)	$f(x)$ is decreasing in the interval (d, ∞)
(III)	$f(x) = -10$ has one real root	(iii)	$\lim_{x \to d^-} f(x) = -\infty$	(R)	$f'(x)$ is increasing in the interval (b, c)
(IV)	$f(x) = 10$ has two real roots	(iv)	$\lim_{x \to \infty} f(x) = 1$	(S)	Local minimum value of $f(x)$ is smaller than local maximum value of $f(x)$

Space for Rough work

44. Which of the following is the only **CORRECT** combination?
(A) $(\mathrm{I})\ (\mathrm{ii})\ (\mathrm{S})$
(B) $(\mathrm{II})\ (\mathrm{iii})\ (\mathrm{Q})$
(C) $(\mathrm{III})\ (\mathrm{iv})\ (\mathrm{S})$
(D) $(\mathrm{IV})\ (\mathrm{iv})\ (\mathrm{S})$

45. Which of the following is the only **INCORRECT** combination?
(A) $(\mathrm{II})\ (\mathrm{ii})\ (\mathrm{P})$
(B) $(\mathrm{III})\ (\mathrm{i})\ (\mathrm{R})$
(C) $(\mathrm{IV})\ (\mathrm{iv})\ (\mathrm{P})$
(D) $(\mathrm{V})\ (\mathrm{iii})\ (\mathrm{Q})$

46. Which of the following is the only **CORRECT** combination?
(A) $(\mathrm{III})\ (\mathrm{iv})\ (\mathrm{P})$
(B) $(\mathrm{I})\ (\mathrm{ii})\ (\mathrm{R})$
(C) $(\mathrm{IV})\ (\mathrm{i})\ (\mathrm{S})$
(D) $(\mathrm{II})\ (\mathrm{iv})\ (\mathrm{S})$

Space for Rough work

Answer questions Q. 47, Q. 48 and Q. 49 by appropriately matching the information given in the three columns of the following table.

Let $f(x)$ be a double differentiable function with continuous second derivative defined on the interval $[0,1]$ satisfying the equation $\int_0^x \sqrt{1-(f''(t))^2}\,dt = \int_0^x f'(t)\,dt \;\forall x \in [0,1]$ *and* $f(0) = f'(0) = 0$. Column-1; contains the information about interval of value of $f(x)$; Column-2; contains information about value of $f(x)$; Column-3: contains information about interval of value of definite integral of $f(x)$; Match the following Column(s)

	Column 1		Column 2		Column 3
(I)	$f\left(\frac{1}{2}\right) < \frac{1}{8}$	(i)	$f\left(\frac{1}{3}\right) = \frac{1}{18}$	(P)	$\int_0^1 f(x)\,dx > 1$
(II)	$f\left(\frac{1}{2}\right) > \frac{1}{8}$	(ii)	$f\left(\frac{1}{4}\right) = \frac{1}{8}$	(Q)	$\int_0^1 f(x)\,dx > \frac{1}{6}$
(III)	$f\left(\frac{1}{3}\right) < \frac{1}{18}$	(iii)	$f\left(\frac{1}{6}\right) = 2\sin^2\left(\frac{1}{12}\right)$	(R)	$\int_0^1 f(x)\,dx < \frac{1}{6}$
(IV)	$f\left(\frac{1}{4}\right) < \frac{1}{4}$	(iv)	$f\left(\frac{1}{8}\right) = -2\sin^2\left(\frac{1}{16}\right)$	(S)	$\int_0^1 f(x)\,dx < \frac{1}{4}$

47. Which of the following is the only **CORRECT** combination?

(A) $(\mathrm{I})\ (\mathrm{ii})\ (\mathrm{Q})$

(B) $(\mathrm{II})\ (\mathrm{i})\ (\mathrm{S})$

(C) $(\mathrm{IV})\ (\mathrm{ii})\ (\mathrm{Q})$

(D) $(\mathrm{III})\ (\mathrm{iii})\ (\mathrm{R})$

Space for Rough work

48. Which of the following is the only **CORRECT** combination?
(A) $(I)\ (iv)\ (Q)$
(B) $(II)\ (iii)\ (S)$
(C) $(II)\ (iv)\ (P)$
(D) $(IV)\ (iii)\ (S)$

49. Which of the following is the only **CORRECT** combination?
(A) $(III)\ (ii)\ (Q)$
(B) $(IV)\ (ii)\ (P)$
(C) $(I)\ (iii)\ (R)$
(D) $(III)\ (ii)\ (S)$

SECTION – B
(Single digit integer type)

This section contains **FIVE** questions. The answer to each question is a single Digit integer ranging from 0 to 9, both inclusive.

50. If p, q, r and s are non – zero numbers such that 'r' and 's' are roots of $x^2 + px + q = 0$ and p, q are the roots of $x^2 + rx + s = 0$, then find the value of $(4p + q + 2r + s)$.

51. Value of $5^{\log_{1/5}(1/2)} + \log_{\sqrt{2}} \frac{4}{\sqrt{7}+\sqrt{3}} + \log_{1/2} \frac{1}{10+2\sqrt{21}}$

52. Let $1^P + 2^P + 3^P + ... + n^P = a_0 n^{P+1} + a_1 n^P + a_2 n^{P-1} ..... + a_P n + a_{P+1}$, where $n, P \in N$. If $a_0 = \frac{1}{8}$, then P = _____

53. $y = f(x)$ is a particular solution of differential equation $\left(\left(\frac{dy}{dx}\right)^2 + y\frac{d^2y}{dx^2}\right) = e^x$ such that $f(0) = 1 = f'(0)$, then the value of $f^2(\ln 2)$ is ……

54. The differential equation $\frac{dy}{dx} = \frac{(x+y+2)^2 - (y-1)^2}{(x+3)^2}$ determines a curve $y = f(x)$ & $f(0) = 1$ then f(3) is equal to …

Space for Rough work